

RRSS	
Risk Management Report	Date: 2024.04.16

RRSS

Risk Management Report

1 Introduction

In software engineering, a risk is any potential event that could negatively impact the development or success of a software product. These risks can manifest in various ways, such as causing delays, exceeding budgets, compromising quality, or leading to security vulnerabilities.

Effectively managing these risks is crucial for ensuring a project's smooth execution and delivering a product that meets its goals. This process typically involves three key stages:

Risk Identification:

This involves brainstorming and actively searching for potential problems that could arise during the project lifecycle. Techniques like brainstorming sessions, analyzing past projects, and considering industry trends can help identify these risks.

Risk Analysis:

Once identified, each risk needs to be carefully assessed. This analysis considers the likelihood (probability) of the risk occurring and the potential impact (severity) it could have on the project. Additionally, factors like the detectability of the risk and the ease of mitigation are evaluated.

Risk Monitoring:

Throughout the project, identified risks need to be continuously monitored. This involves tracking their status, evaluating any changes in likelihood or impact, and ensuring that mitigation strategies remain effective.

This Risk Management Report outlines the potential risks associated with the Review and Recommendation System (RRSS) project. It details these risks in order of importance, along with specific mitigation or contingency actions for each. The following sections delve deeper into the descriptions of these identified risks and the chosen approaches to manage them.

2 Description

The descriptions of the items in the Risk List document.

Section	Definition
Headline	Short description of the risk
Description	Detailed description of the risk
Type	The types of the risks. There are two types: Direct Risks (D): Risks that developers have a direct impact on when making the project. Indirect Risks (I): Risks that developers have not a direct impact on when making the project.

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Impact	Indicates the potential impact on the project as a number between 1-5.
Probability	Indicates the probability of the risk. It is in the range of 1%-100%.
Magnitude	It is a number that shows the magnitude of the risk on the project. It is calculated by multiplying <i>Impact</i> and <i>Probability</i> . It is in the range of 0-5.
Owner	The person who is expected to handle in case of risk.
Mitigation Strategy	It is the strategy planned to be implemented to handle the situation in case of risk.

The table below contains the potential risks specified in the Risk List and the mitigation strategies of those risks.

Potential Risk	Mitigation Strategy
Relied on open source software stops being developed.	Find an equivalent dependency that is well maintained.
Instructors make us use different technology and tools during the development stage.	Learn the new technologies and tools.
Time table of the group members doesn't match, and it causes communication problems.	Find a common free time using scheduling tools such as crab.fit or when2meet
There aren't any expert group members on the Spring Boot.	Learn Spring Boot.
3rd party issue tracker and version control system locks out developers.	Create local git patches and send them via email to a team member who has access to Github.
A severe security flaw arised on one of the software dependencies.	Roll back to an older version which doesn't include the vulnerability.
A group member doesn't finish his documentation task before the deadline.	Don't miss the deadline.
The project's scope could expand beyond what was initially planned, leading to delays and increased costs.	Establish clear boundaries upfront, closely monitor changes, and involve stakeholders in decision-making throughout the project.
Performance bottlenecks arise during high user load.	Employ load balancing technologies.
If the products listed on the platform receive consistently poor reviews, it could affect the platform's reputation.	Implement rigorous quality checks, actively seek customer feedback for improvements, and promptly address any issues raised.
Users might find the user interface unintuitive.	Change the user interface according to the

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	feedback.
Inadequate infrastructure or poor design choices could hinder the platform's ability to handle increased traffic as it grows, leading to performance issues.	Plan for both horizontal and vertical scaling options to accommodate growth.

3 Risk Management Report Specifications

The risk management report is developed to identify potential risks and implement measures to mitigate them. Writing such a report is essential when developing a big software application, as numerous risks can arise during the project development phase. Despite these risks, maintaining uninterrupted progress is crucial for the project's timely completion.

This report includes expected risks and strategies to minimize them. Since it's impossible to foresee every risk, the report might be updated in the future.

The detailed explanations of the risks listed below can be found in the RRSS Risk List table.

Risk ID	Risks forecasted in planning	How it was handled / mitigated
3	Timetable mismatches	Stakeholders scheduled a common free time using crab.fit.
4	Lack of expertise in Spring Boot	Developers learned to work with Spring Boot.
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Risk ID	Risks NOT forecasted in planning, but observed	How it was handled / mitigated
5	Github outages in its 2FA service	The affected developer created local git patches and sent them via email to another developer who still had access to Github.
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